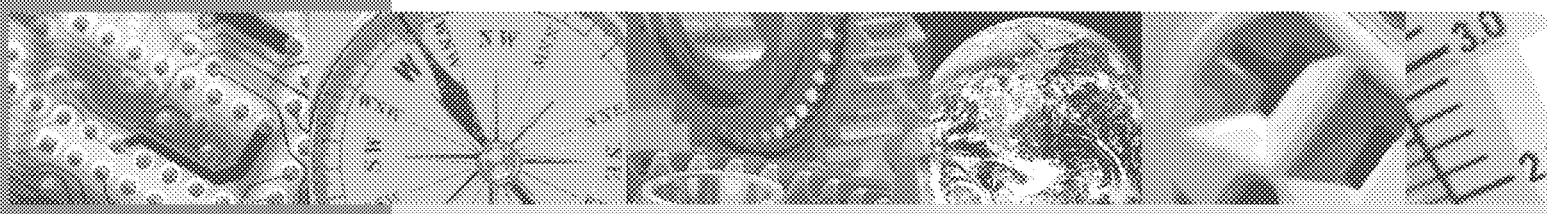
RECOMMENDED PRACTICE

ANSI/ISA–RP12.6–1995



Wiring Practices for Hazardous (Classified) Locations Instrumentation Part 1: Intrinsic Safety

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ANSI/ISA–RP12.6–1995
Wiring Practices for Hazardous (Classified) Locations Instrumentation Part I: Intrinsic Safety

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1 Purpose

1.1* This recommended practice is intended to promote the uniform installation of intrinsically safe systems for hazardous (classified) locations. Information is provided to clarify and explain the requirements of Article 504 of the *National Electrical Code* ® (*NEC* ®).

1.2 This recommended practice applies to the installation of intrinsically safe systems for use in hazardous (classified) locations.

2 Scope

2.1 This recommended practice provides guidance to those who design, install, and maintain intrinsically safe systems for hazardous (classified) locations.

2.2 This recommended practice should be used in conjunction with nationally recognized codes that cover wiring practices — such as the *National Electrical Code* ® (*NEC* ®), ANSI/NFPA 70, and the *Canadian Electrical Code* (*CEC*) Part I, CSA C22.1.

2.3 This recommended practice is not intended to:

a) Include guidance for designing, testing, or repairing intrinsically safe or associated apparatus

b) Apply to the use of portable equipment, except as shown on the control drawing

3 Definitions

For purposes of this recommended practice, the following definitions apply:

3.1 approved:
acceptable to the authority having jurisdiction (*NEC* ®).

3.2 associated apparatus:
(see clause 4.)

3.3 authority having jurisdiction:
the organization, office, or individual that has the responsibility and authority for approving equipment, installations, or procedures.

3.4 channel:
an ungrounded conductor in a grounded intrinsically safe circuit, or a conductor and its reference in a galvanically isolated intrinsically safe circuit.

3.5 control drawing:
(see clause 4.)

3.6 corrective maintenance:
Any maintenance activity that is not normal in the operation of equipment and requires access to the equipment's interior. Such activities are expected to be performed by qualified personnel who are aware of the hazards involved. Such activities typically include locating causes of faulty performance,

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* Further information may be found in annex A.

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replacement of defective components (see 6.2.1) adjustment of internal controls, and the like. Corrective maintenance is referred to simply as maintenance in clause 6.

3.7 different intrinsically safe circuits:
(see clause 4.)

3.8 entity evaluation:
a method used to determine acceptable combinations of intrinsically safe apparatus and connected associated apparatus that have not been investigated in such combination.

3.8.1 Entity parameters for intrinsically safe apparatus:

C_i : The total equivalent internal capacitance that must be considered as appearing across the terminals of the intrinsically safe apparatus.

I_{max} : The maximum DC or peak AC current that can be safely applied to the terminals of the intrinsically safe apparatus. The maximum input current may be different for different terminals.

L_i : The total equivalent internal inductance that must be considered as appearing across the terminals of the intrinsically safe apparatus.

V_{max} : The maximum DC or peak AC voltage that can be safely applied to the terminals of the intrinsically safe apparatus. The maximum input voltage may be different for different terminals.

3.8.2 Entity parameters for associated apparatus:

C_a : The maximum value of capacitance that may be connected to the intrinsically safe circuit of the associated apparatus.

I_{sc} : The maximum DC or peak AC current that may be drawn from the intrinsically safe connections of the associated apparatus.

L_a : The maximum value of inductance that may be connected to the intrinsically safe circuit of the associated apparatus.

V_{oc} : The maximum DC or peak AC open circuit voltage that can appear across the intrinsically safe connections of the associated apparatus.

3.8.3 Additional entity parameters for associated apparatus with multiple channels may include the following:

I_t : The maximum DC or peak AC current that can be drawn from any combination of terminals of a multiple-channel associated apparatus configuration.

V_t : The maximum DC or peak AC open circuit voltage that can appear across any combination of terminals of a multiple-channel associated apparatus configuration.

3.9 galvanic isolation:
the transfer of electrical power or signal from one circuit to another by means that do not include a direct electrical connection — e.g., through an isolating transformer or optical coupler.

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3.10 hazardous (classified) location:
a location where fire or explosion hazards may exist due to the presence of flammable gases or vapors, flammable liquids, combustible dust, or easily ignitible fibers or flyings.

3.11 intrinsic safety:
a type of protection in which a portion of the electrical system contains only intrinsically safe equipment (apparatus, circuits, and wiring) that is incapable of causing ignition in the surrounding atmosphere. No single device or wiring is intrinsically safe by itself (except for battery-operated self-contained apparatus such as portable pagers, transceivers, gas detectors, etc., which are specifically designed as intrinsically safe self-contained devices), but is intrinsically safe only when employed in a properly designed intrinsically safe system. This type of protection is referred to by the International Electrotechnical Commission (IEC) as "Ex i." Also see "associated equipment (apparatus)."

3.12 intrinsic safety barrier:
a network designed to limit the energy (voltage and current) available to the protected circuit in the hazardous (classified) location, under specified fault conditions.

3.13 intrinsic safety ground system:
a grounding system that has a dedicated conductor isolated from the power system, except at one point, so that ground currents will not normally flow and is reliably connected to a grounding electrode in accordance with Article 250 of the *NEC* ® or Section 10 of CEC Part I, CSA C22.1.

3.14 intrinsically safe apparatus:
(see clause 4.)

3.15 intrinsically safe circuit:
(see clause 4.)

3.16 intrinsically safe systems:
(see clause 4.)

3.17 labeled:
equipment or materials to which has been attached a label, symbol, or other identifying mark of an organization acceptable to the authority having jurisdiction and concerned with product evaluation, that maintains periodic inspection of production of labeled equipment or materials, and by whose labeling the manufacturer indicates compliance with appropriate standards or performance in a specified manner. (See *NEC* ® reference.)

3.18 listed:
equipment or materials included in a list published by an organization acceptable to the authority having jurisdiction and concerned with product evaluation, that maintains periodic inspection of production of listed equipment or materials, and by whose listing states that the equipment or material meets appropriate designated standards or has been tested and found suitable for use in a specified manner. (See *NEC* ® reference.)

3.19 nonhazardous location:
a location not designated as hazardous (classified). The term "unclassified location" is also used in the *NEC* ®.

3.20 qualified person:
one familiar with the construction and operation of the equipment and the hazards involved. (See *NEC* ® reference.)

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3.21 simple apparatus:
a device that will neither generate nor store more than 1.2 V, 0.1 A, 25 mW, or 20 μJ; for example: switches, thermocouples, light-emitting diodes, connectors, and resistance temperature devices (RTDs).

3.22 wiring drawing:
a drawing or other document created by the user based upon the relevant control drawings. The wiring drawing is used by the installer to determine the type, color, and size of the wire used to connect each terminal of the equipment used in the intrinsically safe circuit.

4 Article 504 of the NEC (ANSI/NFPA 70-1993) with explanation

NOTE: Throughout clause 4, text that has been excerpted from the *National Electrical Code*® (*NEC*®) is distinguished from the main body of text as follows:

- **NEC® Article 504 text is shaded and indented at the left and right margins.**
- **Other excerpted NEC® text (such as articles on sealing) is shaded but not indented.**

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504-1. Scope. This article covers the installation of intrinsically safe (I.S.) apparatus, wiring, and systems for Class I, II, and III locations.

(FPN): For further information, see Installation of Intrinsically Safe Instrument Systems in Class I Hazardous Locations, ANSI/ISA RP12.6-1987.

Prior to publication of the 1990 *NEC*®, ANSI/ISA-RP12.6, Installation of Intrinsically Safe Systems for Hazardous (Classified) Locations, was the recommended practice for the installation of intrinsically safe systems. The ISA SP12 committee proposed the addition of Article 504 to provide a more enforceable set of requirements for inspection authorities.

504-2. Definitions. For the purpose of this article:

Associated Apparatus: Apparatus in which the circuits are not necessarily intrinsically safe themselves, but that affect the energy in the intrinsically safe circuits and are relied upon to maintain intrinsic safety. Associated apparatus may be either:

1. electrical apparatus that has an alternative type of protection for use in the appropriate hazardous (classified) location, or

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2. electrical apparatus not so protected that shall not be used within a hazardous (classified) location.

(FPN): Associated apparatus has identified intrinsically safe connections for intrinsically safe apparatus and also may have connections for nonintrinsically safe apparatus.

Intrinsic safety barriers are a common form of associated apparatus. These barriers are connected between the intrinsically safe apparatus and the control equipment. Their primary purpose is to limit the energy to the hazardous location under fault conditions. They may also provide isolation, signal conditioning, or both. There are also many types of associated apparatus that normally are not referred to as intrinsic safety barriers, but have energy-limiting circuits suitable for connection directly to intrinsically safe apparatus. An example of this type of associated apparatus is a controller that is not itself intrinsically safe, but has connections for intrinsically safe sensors.

Control Drawing: A drawing or other document provided by the manufacturer of the intrinsically safe or associated apparatus that details the allowed interconnections between the intrinsically safe and associated apparatus.

Different Intrinsically Safe Circuits: Different intrinsically safe circuits are intrinsically safe circuits in which the possible interconnections have not been evaluated and approved as intrinsically safe.

Intrinsically Safe Apparatus: Apparatus in which all the circuits are intrinsically safe.

Intrinsically Safe Circuit: A circuit in which any spark or thermal effect is incapable of causing ignition of a mixture of flammable or combustible material in air under prescribed test conditions.

(FPN): Test conditions are described in Standard for Safety, Intrinsically Safe Apparatus and Associated Apparatus for Use in Class I, II, and III, Division 1, Hazardous (Classified) Locations, ANSI/UL 913-1988.

Intrinsically Safe System: An assembly of interconnected intrinsically safe apparatus, associated apparatus, and interconnecting cables in that those parts of the system that may be used in hazardous (classified) locations are intrinsically safe circuits.

(FPN): An intrinsically safe system may include more than one intrinsically safe circuit.

504-3. Application of Other Articles. Except as modified by this article, all applicable articles of this Code shall apply.

Although intrinsically safe circuits are inherently low energy circuits, they may still be shock hazards because of the operating voltage.

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Clause 500-2 provides an exception for intrinsically safe apparatus and wiring from the requirements of Articles 501 through 503 and 510 through 516. All other articles of the Code apply to intrinsically safe wiring, except as exempted by specific articles.

If the rated voltage of the circuit exceeds 60 volts DC or 30 volts AC, the wiring requirements for Class 3 circuits apply. (See *NEC* ® Article 725.)

Other articles may apply, depending on the functional application — e.g., Article 760 for fire protective signaling systems, Article 800 for communications circuits, and clause 725-49 for cables installed in ducts, plenums, risers, and other air-handling spaces.

504-4. Equipment Approval. All intrinsically safe apparatus and associated apparatus shall be approved.

Electrical equipment that is listed or labeled by a nationally recognized testing laboratory (NRTL) normally will be accepted by the authority having jurisdiction. The authority having jurisdiction may also accept specialized equipment not listed or labeled by an NRTL, with appropriate technical justification. A written report of the investigation and conclusion should be kept on file, and the markings on the equipment should identify the report.

504-10. Equipment Installation.

(a) Control Drawing. Intrinsically safe apparatus, associated apparatus, and other equipment shall be installed in accordance with the control drawing(s).

(FPN): The control drawing identification is marked on the apparatus.

There are three basic types of control drawings:

a) Intrinsically safe apparatus and associated apparatus are specified by manufacturer and model number. (See figure 4.1 for an example.)

b) Intrinsically safe apparatus is specified by manufacturer and model number for connection to associated apparatus specified by entity parameters. (See figure 4.2 for an example).

c) Associated apparatus is specified by manufacturer and model number for connection to intrinsically safe apparatus that is specified by entity parameters or to simple apparatus. (See figure 4.3 for an example.)

Control drawings that are combinations of the above types are also possible. For example, control drawings for intrinsically safe apparatus often specify permissible connections to specific associated apparatus and also specify entity parameters to allow additional flexibility in selecting associated apparatus.

To ensure that a given interconnection forms an intrinsically safe system, it is necessary to obtain control drawings that specify each intrinsically safe apparatus and associated apparatus to be interconnected. If a control drawing of the type shown in figure 4.1 that correctly describes the interconnection is available, only that control drawing is necessary.

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If the intrinsic safety of the system is to be based on the comparison of entity parameters, it is necessary to obtain a control drawing for each intrinsically safe apparatus and associated apparatus. Care should be taken to ensure that the entity parameters used in the comparison apply to the specific set(s) of terminals to be interconnected.

If the system includes only simple apparatus connected to an associated apparatus, only the associated apparatus control drawing is necessary. Multiple channels of associated apparatus should not be connected to a single simple apparatus unless specifically permitted by the control drawing.

Frequently, the user creates a wiring drawing based on the control drawings provided by the manufacturers of the intrinsically safe apparatus and associated apparatus or other specification sheets that provide information such as terminal identification.

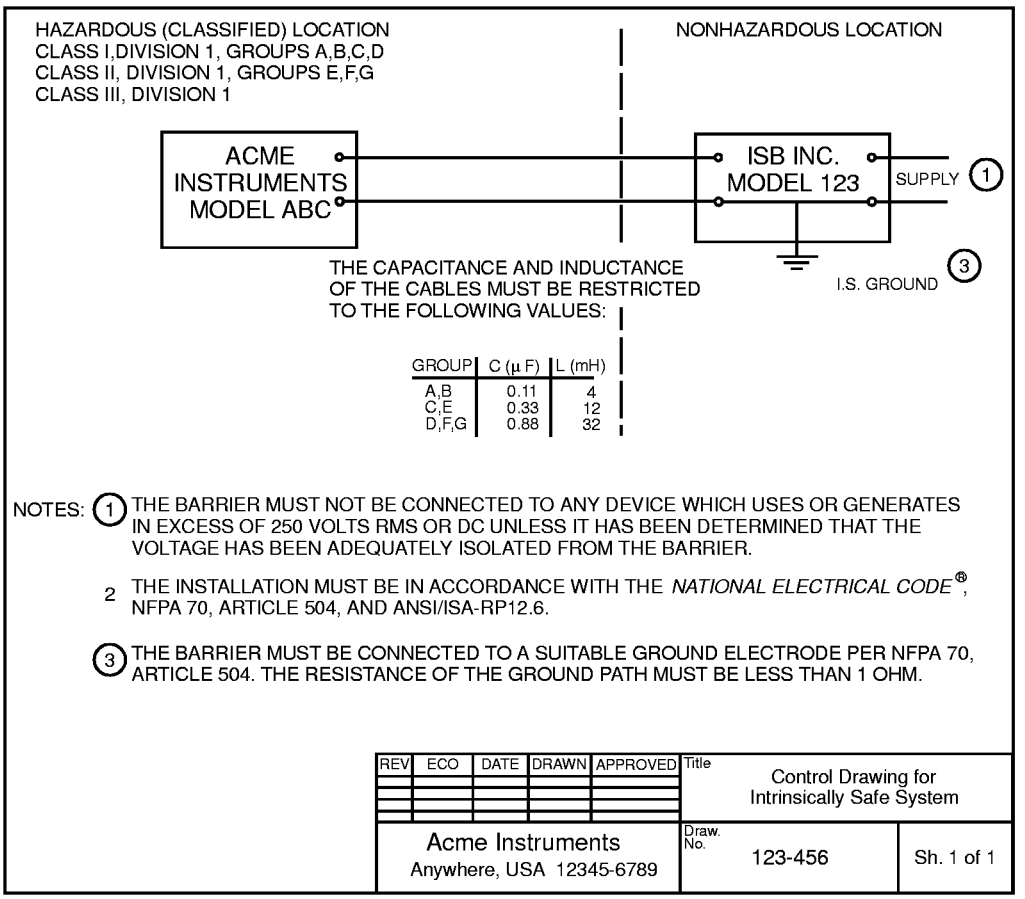


Figure 4.1 — Example of a control drawing for an intrinsically safe system

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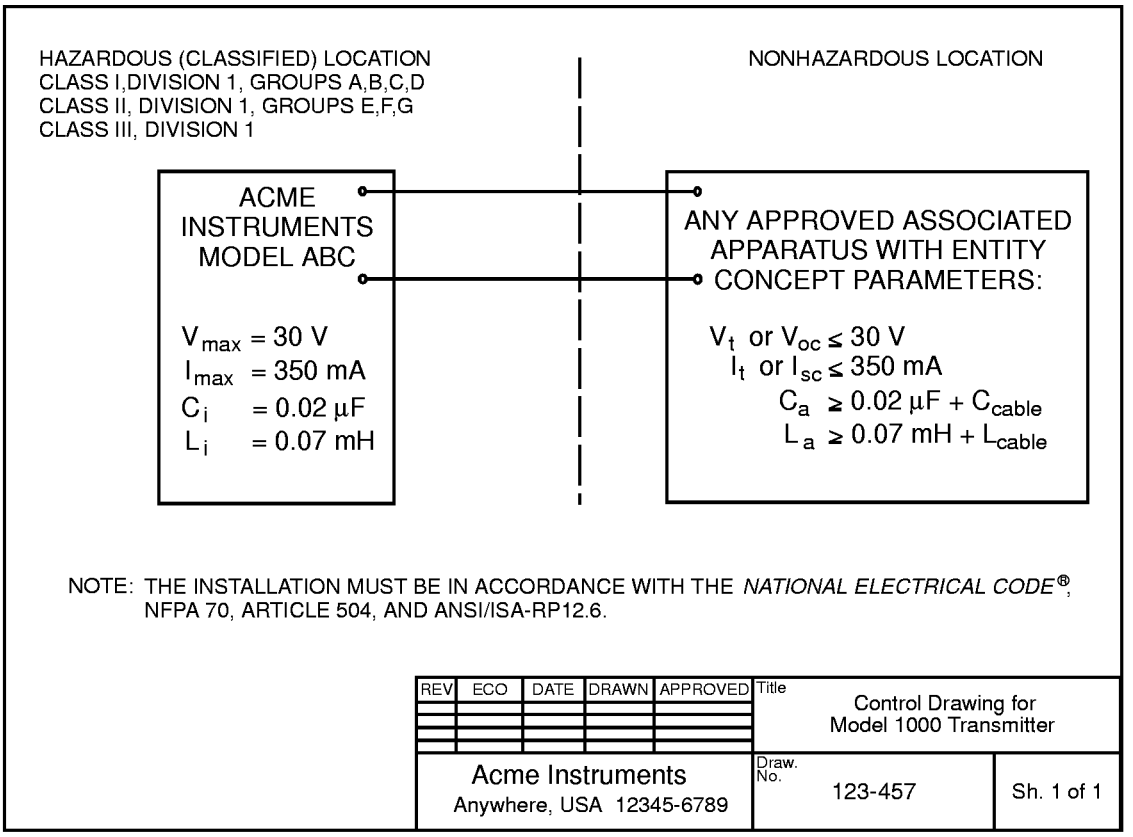
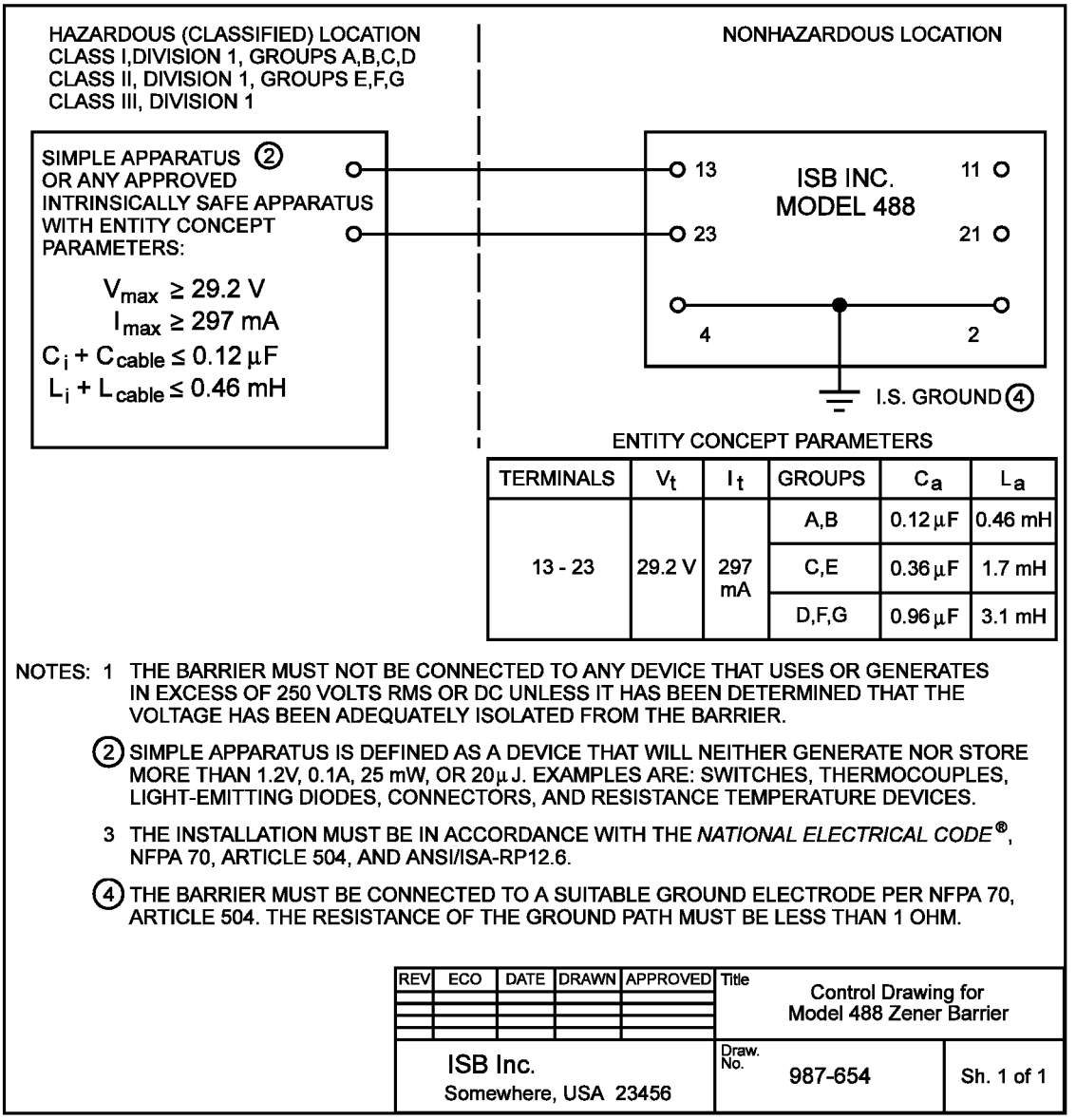


Figure 4.2 — Example of control drawing for an intrinsically safe apparatus with entity parameters

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ENTITY CONCEPT PARAMETERS

TERMINALS	V_t	I_t	GROUPS	C_a	L_a
13 - 23	29.2 V	297 mA	A,B	0.12 µF	0.46 mH
			C,E	0.36 µF	1.7 mH
			D,F,G	0.96 µF	3.1 mH

NOTES: 1 THE BARRIER MUST NOT BE CONNECTED TO ANY DEVICE THAT USES OR GENERATES IN EXCESS OF 250 VOLTS RMS OR DC UNLESS IT HAS BEEN DETERMINED THAT THE VOLTAGE HAS BEEN ADEQUATELY ISOLATED FROM THE BARRIER.

② SIMPLE APPARATUS IS DEFINED AS A DEVICE THAT WILL NEITHER GENERATE NOR STORE MORE THAN 1.2V, 0.1A, 25 mW, OR 20µ J. EXAMPLES ARE: SWITCHES, THERMOCOUPLES, LIGHT-EMITTING DIODES, CONNECTORS, AND RESISTANCE TEMPERATURE DEVICES.

3 THE INSTALLATION MUST BE IN ACCORDANCE WITH THE *NATIONAL ELECTRICAL CODE®*, NFPA 70, ARTICLE 504, AND ANSI/ISA-RP12.6.

④ THE BARRIER MUST BE CONNECTED TO A SUITABLE GROUND ELECTRODE PER NFPA 70, ARTICLE 504. THE RESISTANCE OF THE GROUND PATH MUST BE LESS THAN 1 OHM.

REV	ECO	DATE	DRAWN	APPROVED	Title
					Control Drawing for Model 488 Zener Barrier

ISB Inc. Somewhere, USA 23456	Draw. No. 987-654	Sh. 1 of 1

Figure 4.3 — Example of control drawing for an associated apparatus with entity parameters

(b) **Location:** Intrinsically safe and associated apparatus shall be permitted to be installed in any hazardous (classified) location for which it has been approved.

(FPN): Associated apparatus may be installed in hazardous (classified) locations if protected by other means permitted by Articles 501 through 503.

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An intrinsically safe system consists of associated apparatus in a nonhazardous or a Division 2 location that is connected by wiring to intrinsically safe apparatus in a Division 1 or Division 2 location. Alternatively, the intrinsically safe circuit may originate in associated apparatus suitable for, and located in, a Division 1 location.

Equipment that has been approved for a Division 1 location may be used in a Division 2 location of the same class and group. (See paragraph 500-3 (a) of the NEC®.)

Some examples of intrinsically safe systems are given in figure A.1.

General-purpose enclosures shall be permitted for intrinsically safe apparatus.

Intrinsically safe apparatus should be provided with an enclosure that is suitable for the environmental conditions to which it will be exposed (such as temperature, moisture, and corrosion).

504-20. Wiring Methods. Intrinsically safe apparatus and wiring shall be permitted to be installed using any of the wiring methods suitable for unclassified locations. Sealing shall be as provided in 504-70, and separation shall be as provided in 504-30.

Intrinsically safe circuits need not comply, for example, with Articles 501 through 503 and 510 through 516 of the NEC® (1993) or Rules 18-100 through 18-130 of the CEC (1990) and, in general, may be wired in the same manner as comparable circuits intended for use in nonhazardous locations. Examples are PLTC cable in cable trays, nonmetallic cables, and communication cables. Since the energy in an intrinsically safe circuit is inherently limited, no additional overcurrent protection is required in such circuits.

Additional precautions should be taken to provide mechanical protection in applications involving vibration, motion, impacts, etc.

When intrinsically safe wiring may be exposed to disturbing electromagnetic fields, suitable attention should be given to twisting or shielding conductors, or other methods to prevent the energy level of the intrinsically safe wiring from becoming ignition-capable.

504-30. Separation of Intrinsically Safe Conductors.

(a) From Nonintrinsically Safe Circuit Conductors.

(1) Open wiring. Conductors and cables of intrinsically safe circuits not in raceways or cable trays shall be separated at least 2 inches (50 mm) and secured from conductors and cables of any nonintrinsically safe circuits.

Exception: Where either: (1) all of the intrinsically safe circuit conductors are in Type MI, MC, or SNM cables or (2) all of the nonintrinsically safe circuit conductors are in raceways or Type MI, MC, or SNM cables where the sheathing or cladding is capable of carrying fault current to ground.

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(2) In raceways, cable trays, and cables. Conductors of intrinsically safe circuits shall not be placed in any raceway, cable tray, or cable with conductors of any nonintrinsically safe circuit.

Exception No. 1: Where conductors of intrinsically safe circuits are separated from conductors of nonintrinsically safe circuits by a distance of at least 2 inches (50 mm) and secured or by a grounded metal partition or an approved insulating partition.

Braided or aluminum/polyester shielding is not considered suitable for a grounded metal partition. Cable jackets normally are not considered suitable for an insulating partition.

(FPN): No. 20 gauge sheet metal partitions 0.0359 inch (912 micrometers) or thicker are generally considered acceptable.

Exception No. 2: Where either: (1) all of the intrinsically safe circuit conductors, or (2) all of the nonintrinsically safe circuit conductors are in grounded metal-sheathed or metal-clad cables where the sheathing or cladding is capable of carrying fault current to ground.

(FPN): Cables meeting the requirements of Articles 330, 334, and 337 are typical of those considered acceptable.

(3) Within enclosures.

a. Conductors of intrinsically safe circuits shall be separated at least 2 inches (50 mm) from conductors of any nonintrinsically safe circuits or as specified in Section 504-30 (a) (2).

b. All conductors shall be secured so that any conductor that might come loose from a terminal cannot come in contact with another terminal.

(FPN No. 1): The use of separate wiring compartments for the intrinsically safe and nonintrinsically safe terminals is the preferred method of complying with this requirement.

(FPN No. 2): Physical barriers such as grounded metal partitions or approved insulating partitions or approved restricted access wiring ducts separated from other such ducts by at least 3/4 inch (19 mm) can be used to help assure the required separation of the wiring.

Care shall be taken in the layout of terminals and the wiring methods used to prevent contact between intrinsically safe and nonintrinsically safe circuits. Some layouts — e.g., when terminals arranged one

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above another — do not provide adequate separation if a wire should become disconnected. In these cases, additional precautions (such as tie-downs) are necessary.

Clearance between ungrounded terminals and grounded metal should be at least 3 mm (0.125 in.).

A partition may be used to segregate terminals and should extend close enough to the enclosure walls to effectively separate the wiring on either side of the partition. Alternatively, the partition need only extend far enough beyond the terminals to provide 50 mm (2 in.) spacing between intrinsically safe and nonintrinsically safe terminals if the wiring is secured to maintain the required separation.

When several devices having both intrinsically safe and nonintrinsically safe terminals are mounted in the same enclosure, attention must be given to the separation of circuits. An acceptable method of separation is shown in figure A-2. Separate wireways are often used to provide greater assurance that separation of wiring will be maintained. Wire lacing, wire ties, or equivalent fasteners are also acceptable methods of maintaining the 50 mm (2 in.) separation.

Plug-and-socket connectors used to connect intrinsically safe circuits in a nonhazardous location either should not be interchangeable with any other plugs or sockets or should be identified in a way that minimizes the possibility of such interchange.

(b) From Different Intrinsically Safe Circuit Conductors. Different intrinsically safe circuits shall be in separate cables or shall be separated from each other by one of the following means:

(1) The conductors of each circuit are within a grounded metal shield;

(2) The conductors of each circuit have insulation with a minimum thickness of 0.01 inch (254 micrometers).

Exception: Unless otherwise approved.

Clearance between terminals for the connection of different intrinsically safe circuits should be at least 6 mm (0.25 in).

504-50 Grounding.

(a) Intrinsically Safe Apparatus, Associated Apparatus, and Raceways. Intrinsically safe apparatus, associated apparatus, cable shields, enclosures and raceways, if of metal, shall be grounded.

(FPN): Supplementary bonding to the grounding electrode may be needed for some associated apparatus, -- e.g., zener diode barriers, if specified in the control drawing. See Installation of Intrinsically Safe Instrument Systems in Class I Hazardous Locations, ANSI/ISA RP12.6-1987.

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The integrity of a shunt diode intrinsic safety barrier depends on the effective shunting of the ignition-capable electrical current back to the source (to ground).

It is the intent of the following recommendations to ensure that the methods used to connect barriers to ground provide a high integrity, low-resistance return path to the source of the fault current. A separate insulated connection to a grounding electrode will minimize fault currents from other equipment elevating the I.S. ground. Careful consideration should be given to the grounding electrode system(s) to which potential sources of supply and intrinsically safe apparatus are connected. This will enable a determination of whether shunt diode barriers are appropriate (see figure 4.7) and, if so, selection of a grounding electrode.

Exception: The equipment grounding conductor may be used as the intrinsic safety grounding conductor only if potential ground fault current from other equipment that is sharing the AC grounding conductor will not cause an unsafe voltage differential between the grounding electrode and a grounded conductor of an intrinsically safe circuit. Examples of installations not requiring a separate intrinsic safety grounding conductor may include flowmeters with intrinsically safe transducers, consoles with intrinsically safe keyboards, and recorders with intrinsically safe inputs where there is an equipotential bond between the barrier ground and grounded metal parts that the intrinsically safe circuit may contact.

The barrier-grounding terminal must be connected to the grounding electrode. Where there are multiple barriers, the individual grounding terminals may be collected at a common point such as a barrier bus (see figures 4.4 through 4.6). The common point or the grounding terminal on a single barrier must be connected to the grounding electrode using an insulated conductor no smaller than 12 AWG (American Wire Gauge). The wires between individual barriers and the common point may be smaller than 12 AWG. The conductor to the grounding electrode should be identified at both ends to differentiate it from other ground conductors. The conductor must be protected from damage as required by *NEC* ® 250-95(c).

All grounding path connections should be secure, permanent, visible, and accessible. The grounding path resistance from the farthest barrier to the grounding electrode should not exceed 1 ohm.

More than one barrier bus may use the same grounding conductor(s), provided the buses are interconnected in such a way that disconnection of one barrier bus does not result in loss of ground to the other buses.

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Figure 4.4 shows a grounding system in which a separate intrinsic safety ground conductor is connected directly between the barrier bus and the grounding electrode.

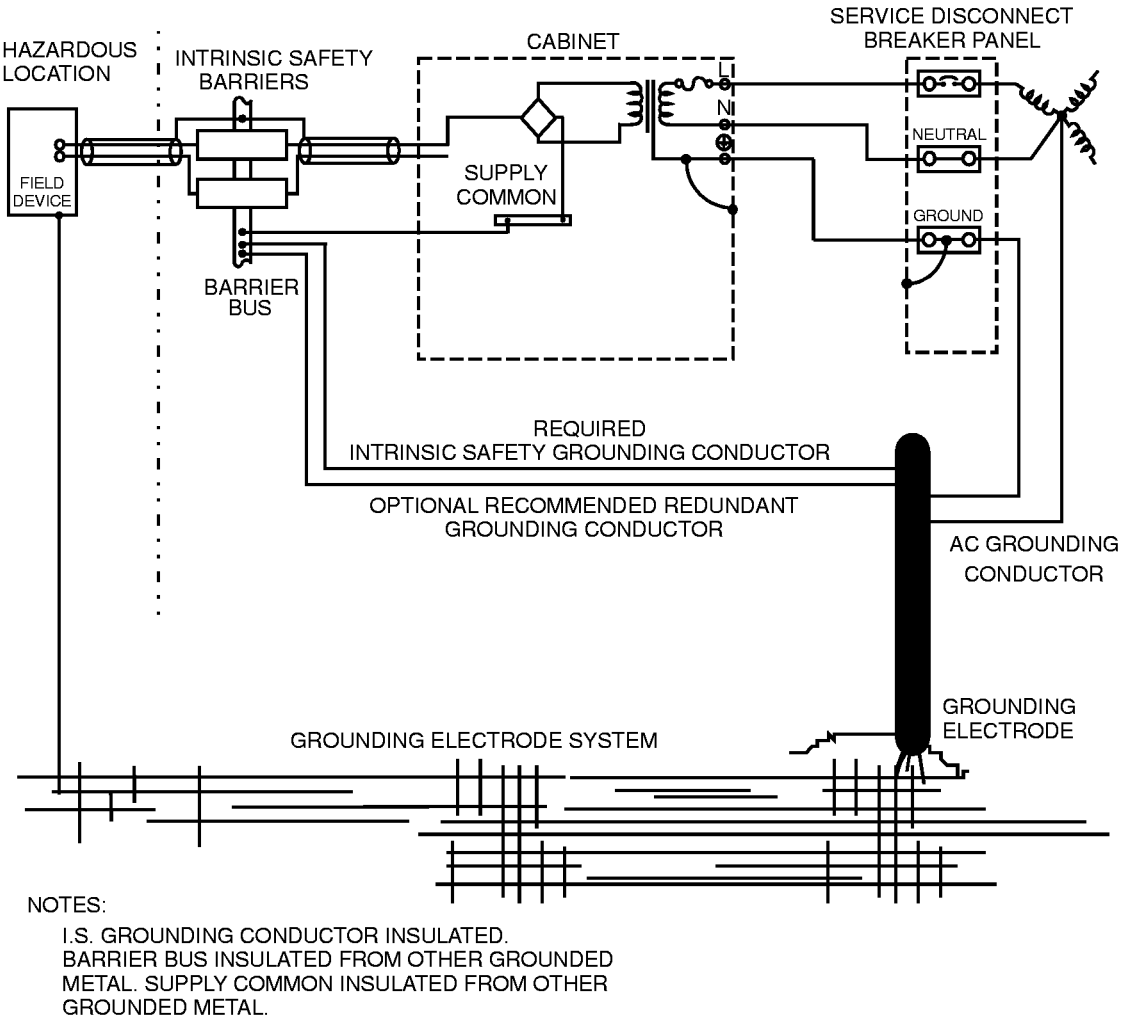


Figure 4.4 — Separate intrinsic safety grounding conductor with field device bonded to same grounding electrode system

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Figure 4.5 shows an alternate grounding system in which the separate intrinsic safety ground conductor is connected between the supply common bus and the grounding electrode.

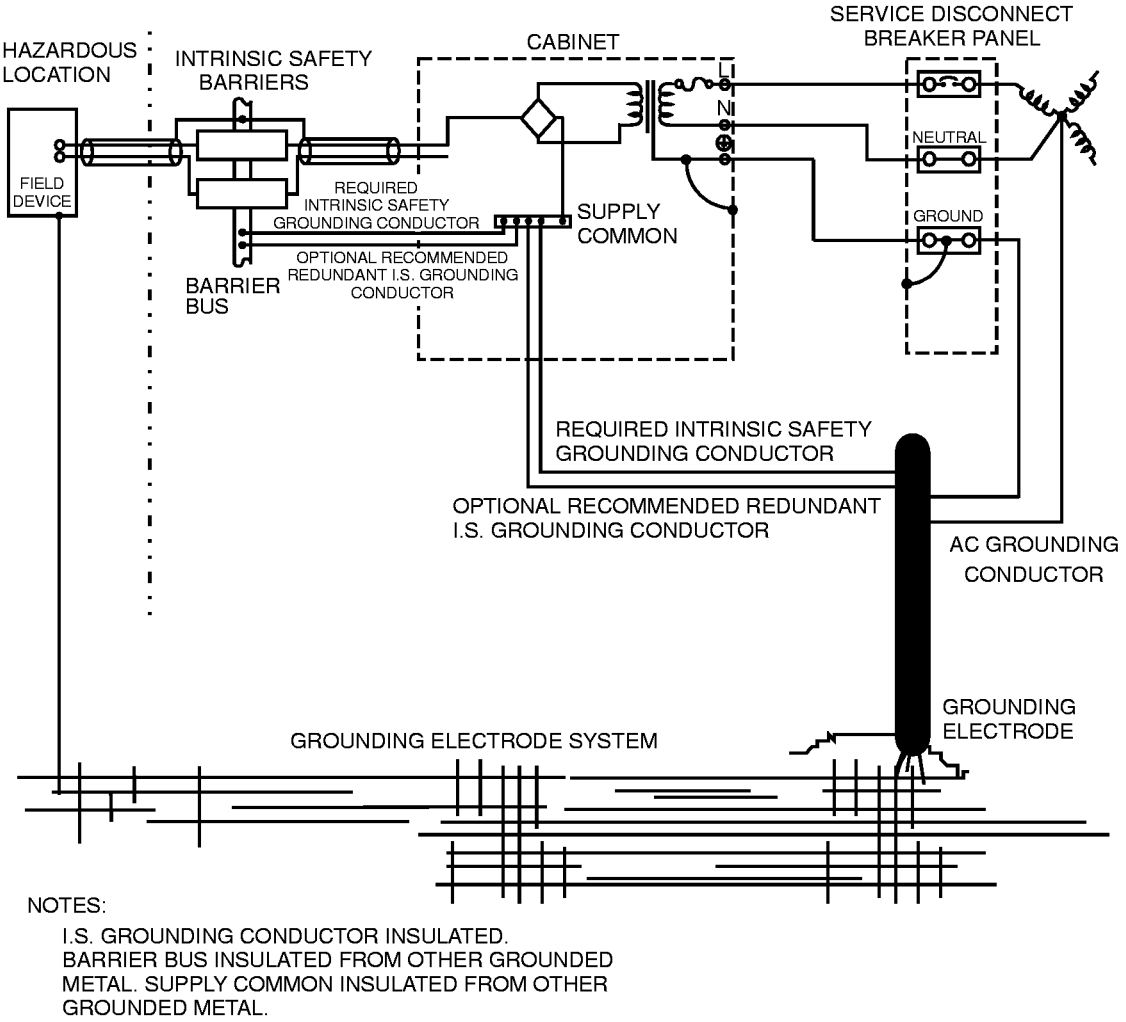


Figure 4.5 — Alternate separate intrinsic safety grounding conductor with field devices bonded to same grounding electrode system

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Figure 4.6 shows an alternate grounding system in which the supply common bus and the barrier bus are connected to a separate master barrier bus bar that is used to interconnect the barrier buses from several cabinets.

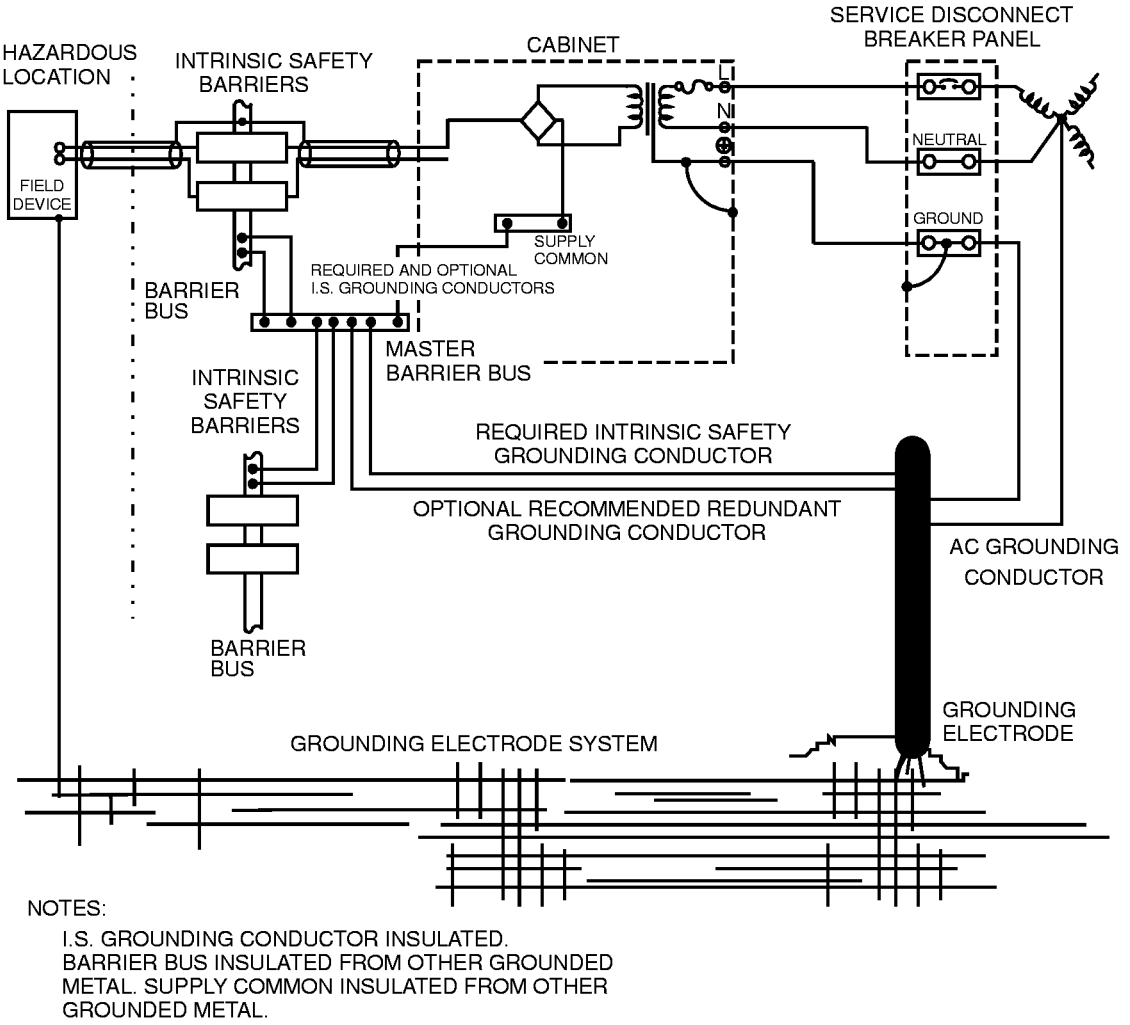


Figure 4.6 — Separate intrinsic safety grounding conductor with field device bonded to same grounding electrode system

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Figure 4.7 shows the need for isolating barriers if the field device is connected to a grounding electrode system different from that used for the control equipment.

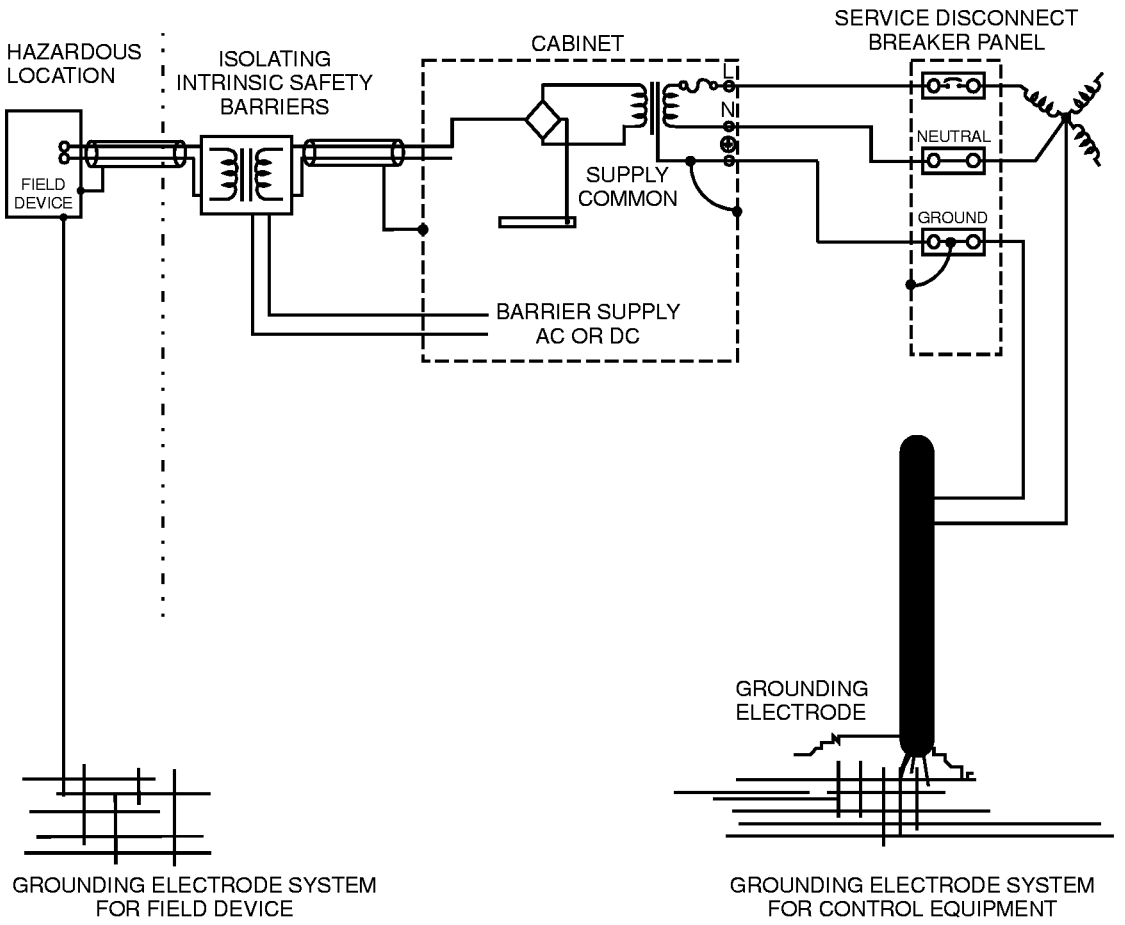


Figure 4.7 — Isolating barrier used. These barriers do not require grounding. Field device is not bonded to same grounding electrode system.

The integrity of the grounding system is essential to maintain the intrinsic safety provided by the shunt diode barriers. In appendix F of the CEC (1990) it is recommended that duplicate grounding conductors be used to connect the shunt diode barriers to the grounding electrode. The use of redundant grounding conductors simplifies measuring the resistance between the grounding electrode and the barrier.

Aluminum conductors should not be used in an intrinsic safety grounding system unless precautions are taken to prevent corrosion at the connection points.

> **(b) Connection to Grounding Electrodes.** Where connection to a grounding electrode is required, the grounding electrode shall be as specified in Sections 250-81(a), (b), (c), and (d) and shall comply with Section 250-26(c). Section 250-83 shall not be used if electrodes specified in Section 250-81 are available.

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(c) Shields. Where shielded conductors or cables are used, shields shall be grounded.

Exception: Where a shield is part of an intrinsically safe circuit.

A shield that is continuous between control equipment and the I.S. apparatus must be at ground potential (see figure 4.8) or connected through associated apparatus (see figure 4.12). If the shield is interrupted at the intrinsic safety barrier, the separate shields may be connected to enclosure ground, chassis ground, or other reference, as performance requirements dictate (see figures 4.9 through 4.11). When connected as in figure 4.12, the V_{oc} and I_{sc} ratings for the barrier connected to the shield must be included in the V_t and I_t assessment. Shields should also be insulated to prevent unwanted ground connections that would conflict with figures 4.8 through 4.12.

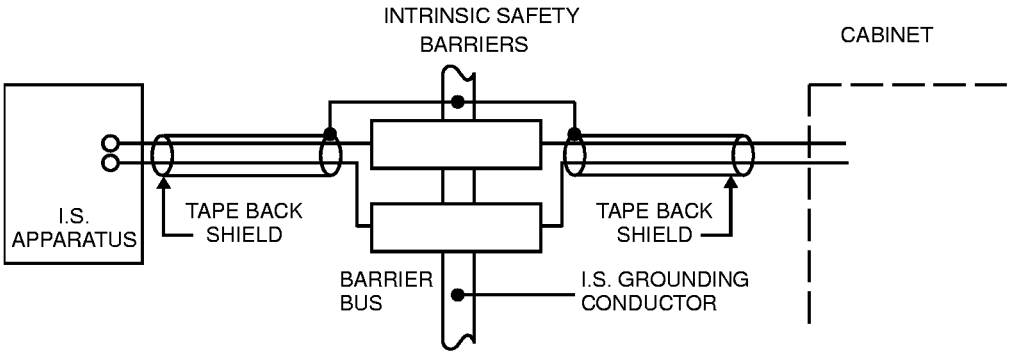


Figure 4.8 — Preferred bonding of shields

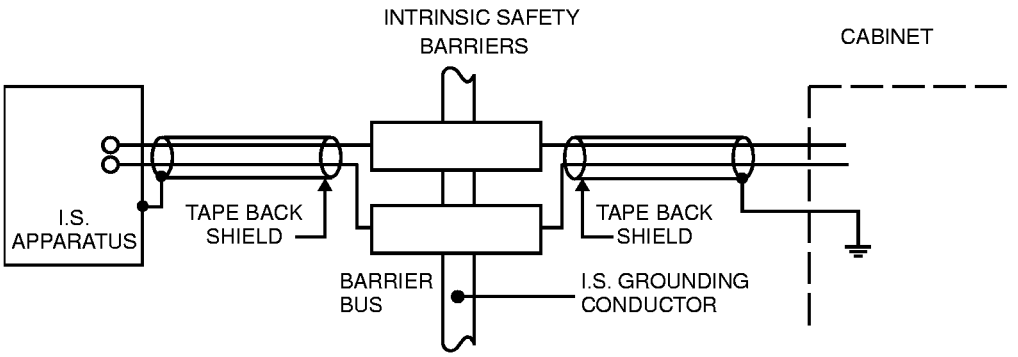


Figure 4.9 — Shield bonding isolated across barrier

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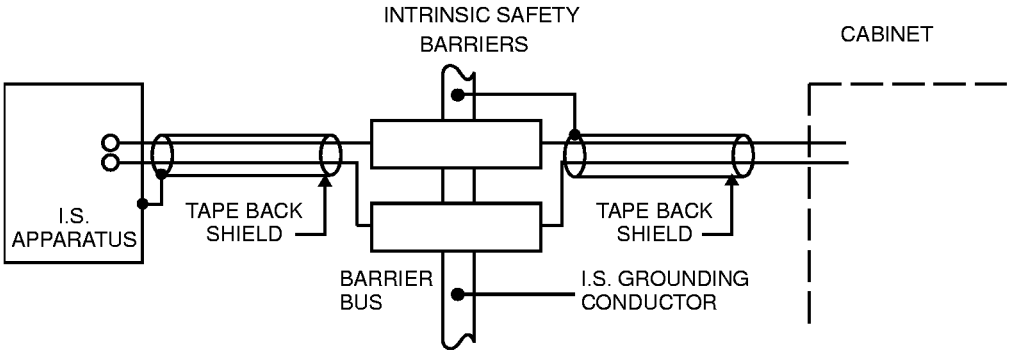


Figure 4.10 — Shield bonding isolated across barrier

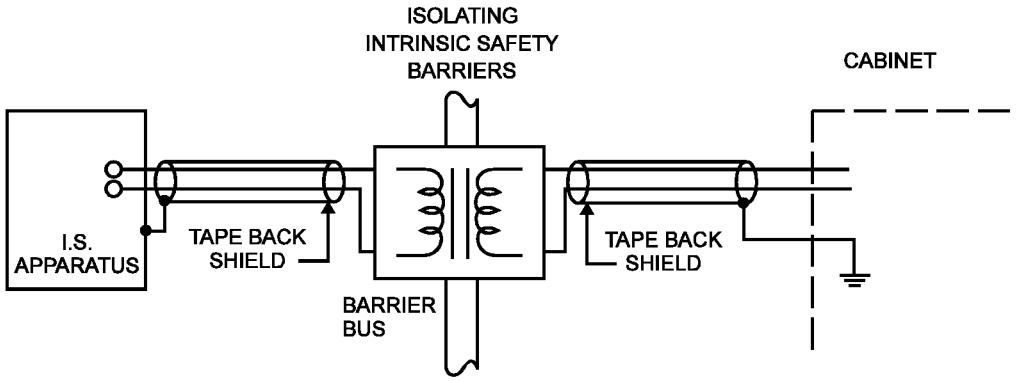


Figure 4.11 — Shields taped back at isolating barrier

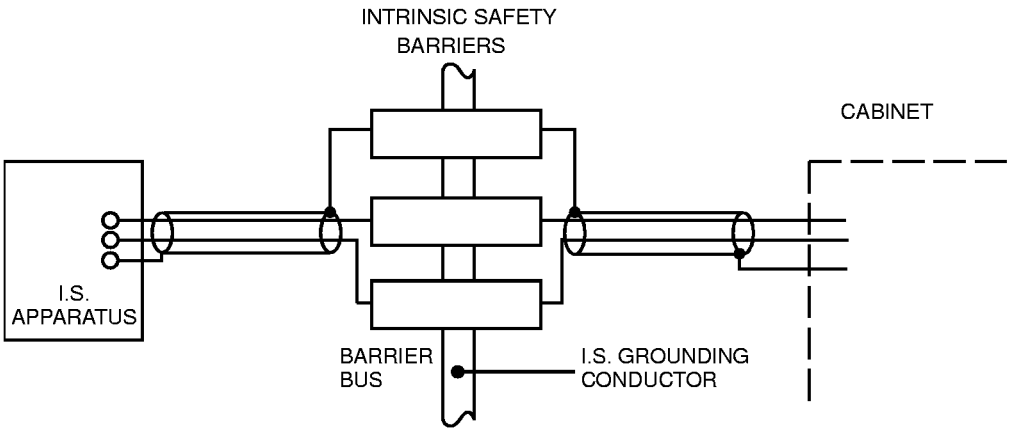


Figure 4.12 — Driven shield using third barrier

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504-60. Bonding.

(a) Hazardous Locations. In hazardous (classified) locations, intrinsically safe apparatus shall be bonded in the hazardous (classified) location in accordance with Section 250-78.

250-78 Bonding in Hazardous (Classified) Locations. Regardless of the voltage of the electrical system, the electrical continuity of noncurrent-carrying metal parts of equipment, raceways, and other enclosures...shall be assured by any of the methods specified for services in Section 250-72 (b) through (e) that are approved for the wiring method used.

250-72 (b) Threaded Connections. Connections utilizing threaded couplings or threaded bosses on enclosures shall be made up wrenchtight where rigid metal conduit or intermediate metal conduit is involved.

250-72 (c) Threadless Couplings and Connectors. Threadless couplings and connectors made up tight for rigid metal conduit, intermediate metal conduit, and electrical metallic tubing. Standard locknuts or bushings shall not be used for the bonding required by this section.

250-72 (d) Bonding Jumpers. Bonding jumpers meeting the other requirements of this article shall be used around concentric or eccentric knockouts that are punched or otherwise formed so as to impair the electrical connection to ground.

250-72 (e) Other Devices. Other approved devices, such as bonding-type locknuts and bushings.

When metal conduit is not used for intrinsically safe circuits, bonding of exposed metal parts must be accomplished through other means, such as bonding conductors.

(b) Nonhazardous Locations. In nonhazardous locations where metal raceways are used for intrinsically safe system wiring in hazardous locations, associated apparatus shall be bonded in accordance with Sections 501-16(a), 502-16(a) or 503-16(a), as applicable.

It is necessary that all raceways, enclosures, etc. located between a hazardous location and the point of grounding are bonded in a fashion similar to the raceways utilized in the hazardous location. The main purpose of the bonding is to provide a low resistance path to ground, to prevent sparking or arcing, in the hazardous location. For example, during a ground fault condition in the associated apparatus enclosure or in the raceway between the enclosure and the power source, this raceway is expected to carry the fault current to its source until the overcurrent device functions to clear the fault. However, if this raceway bonding is a greater resistance than the bonding in the hazardous location, the fault current will flow in the hazardous location. During this interval, some of the current will try to flow through incidental contacts (piping, metal beams, etc.) Since those incidental contacts, in the hazardous location, may not be able to handle such fault currents, a spark, arc, or heated metal could result.

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504-70. Sealing. Conduits and cables that are required to be sealed by clauses 501-5 and 502-5 shall be sealed to minimize the passage of gases, vapors or dust.

Exception: Seals are not required for enclosures that contain only intrinsically safe apparatus except as required by section 501-5(f)(3).

(FPN): It is not the intent of this section to require an explosionproof seal.

The following sections pertain to sealing and drainage of intrinsically safe systems. If an explosionproof installation is required, explosionproof fittings are required for the intrinsically safe circuits leaving the enclosure.

501-5 Sealing and Drainage. Seals in conduit and cable systems shall comply with (a) through (f) below. Sealing compound shall be of a type approved for the conditions and use. Sealing compound shall be used in Type MI cable termination fittings to exclude moisture and other fluids from the cable insulation.

(FPN No. 1): Seals are provided in conduit and cable systems to minimize the passage of gases and vapors and prevent the passage of flames from one portion of the electrical installation to another through the conduit. Such communication through Type MI cable is inherently prevented by construction of the cable. Unless specifically designed and tested for the purpose, conduit and cable seals are not intended to prevent the passage of liquids, gases, or vapors at a continuous pressure differential across the seal. Even at differences in pressure across the seal equivalent to a few inches of water, there may be a slow passage of gas or vapor through a seal and through conductors passing through the seal. See clause 501- 5(e)(2). Temperature extremes and highly corrosive liquids and vapors can affect the ability of seals to perform their intended function. See clause 501-5(c)(2).

(FPN No. 2): Gas or vapor leakage and propagation of flames may occur through the interstices between the strands of standard stranded conductors larger than No. 2. Special conductor constructions e.g., compacted strands or sealing of the individual strands, are means of reducing leakage and preventing the propagation of flames.

(a) Conduit Seals, Class I, Division 1. In Class I, Division 1 locations, conduit seals shall be located as follows:

Paragraphs 501-5(a)(1), 501-5(a)(2), and 501-5(a)(3) do not apply to equipment containing only intrinsically safe circuits.

(4) In each conduit run leaving the Class I, Division 1 location. The sealing fitting shall be permitted on either side of the boundary of such location but shall be so designed and installed to minimize the amount of gas or vapor that may have entered the conduit system with the Division 1 location from being communicated to the conduit beyond the seal. There shall be no union, coupling, box or fitting in the conduit between the sealing fitting and the point at which the conduit leaves the Division 1 location.

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Exception: Metal conduit containing no unions, couplings, boxes, or fittings that passes completely through a Class I, Division 1 location with no fittings less than 12 inches (305 mm) beyond each boundary shall not be required to be sealed if the termination points of the unbroken conduit are in unclassified locations.

(b) Conduit Seals, Class I, Division 2. In Class I, Division 2 locations, conduit seals shall be located as follows:

Paragraph 501-5 (b)(1) does not apply to intrinsically safe apparatus.

(2) In each conduit run passing from a Class I, Division 2 location into an unclassified location. The sealing fitting shall be permitted on either side of the boundary of such a location but shall be so designed and installed to minimize the amount of gas or vapor that may have entered the conduit system within the Division 2 location from being communicated to the conduit beyond the seal. Rigid metal conduit or threaded steel intermediate metal conduit shall be used between the sealing fitting and the point at which the conduit leaves the Division 2 location, and a threaded connection shall be used at the sealing fitting. There shall be no union, coupling, box, or fitting in the conduit between the sealing fitting and the point at which the conduit leaves the Division 2 location.

Exception No. 1: Metal conduit containing no unions, couplings, boxes, or fittings that passes completely through a Class I, Division 2 location with no fittings less than 12 inches (305 mm) beyond each boundary shall not be required to be sealed if the termination points of the unbroken conduit are in unclassified locations.

Exception No. 2: Conduit systems terminating at an outdoor unclassified location where a wiring method transition is made to cable tray, cablebus, ventilated busway, TYPE MI cable, or open wiring shall not be required to be sealed where passing from the Class I, Division 2 location into the unclassified area. The conduits shall not terminate at an enclosure containing an ignition source.

(c) Class I, Divisions 1 and 2. Where required, seals in Class I, Division 1 and 2 locations shall comply with the following:

(1) Fittings. Enclosures for connections or equipment shall be provided with an approved integral means for sealing, or sealing fittings approved for Class I locations shall be used. Sealing fittings shall be accessible.

(2) Compound. Sealing compound shall be approved and shall provide a seal against passage of gas or vapors through the seal fitting, shall not be affected by the surrounding atmosphere or liquids and shall not have a melting point of less than 93°C (200°F).

(3) Thickness of compounds. In a completed seal, the minimum thickness of the sealing compound shall not be less than the trade size of the conduit and in no case less than 5/8 inch (16 mm).

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(4) Splices and taps. Splices and taps shall not be made in fittings intended only for sealing with compound, nor shall other fittings in which splices or taps are made be filled with compound.

(5) Assemblies. In an assembly where equipment that may produce arcs, sparks, or high temperatures is located in a compartment separate from the compartment containing splices or taps, and an integral seal is provided where conductors pass from one compartment to the other, the entire assembly shall be approved for Class I locations. Seals in conduit connections to the compartment containing splices or taps shall be provided in Class I, Division 1 locations where required by (a)(2) above.

(d) Cable seals, Class I, Division 1. In Class I, Division 1 locations each multiconductor cable in conduit shall be considered as a single conductor if the cable is incapable of transmitting gases or vapors through the cable core. These cables shall be sealed in accordance with (a) above.

Cable with a gas/vapor-tight continuous sheath capable of transmitting gases or vapors through the cable core shall be sealed in the Division 1 location after removing the jacket and any other coverings so that the sealing compound will surround each individual insulated conductor and the outer jacket.

Exception: Multiconductor cables with a gas/vapor-tight continuous sheath capable of transmitting gases or vapors through the cable core shall be permitted to be considered as a single conductor by sealing the cable in the conduit within 18 inches (457 mm) of the enclosure and the cable end within the enclosure by an approved means to prevent the entrance of gases or vapors or propagation of flame into the cable core, or by other approved methods.

Cables not installed in conduit are permitted for intrinsically safe circuits, but the above rules do not cover the sealing requirements. Refer to the NEC® or CEC, as applicable, for Division 2 requirements for sealing cables not in conduit.

(e) Cable Seals, Class I, Division 2. In Class I, Division 2 locations, cable seals shall be located as follows:

Paragraph 501-5 (e)(1) does not apply to intrinsically safe apparatus.

(2) Cables with a gas/vapor-tight continuous sheath and which will not transmit gases or vapors through the cable core in excess of the quantity permitted for seal fittings shall not be required to be sealed... The minimum length of such cable run shall not be less than that length which limits gas or vapor flow through the cable core to the rate permitted for seal fittings [0.007 cubic feet per hour (198 cubic centimeters per hour) of air at a pressure of 6 inches of water (1493 pascals).]

(3) Cables with a gas/vapor-tight continuous sheath capable of transmitting gases or vapors through the cable core shall not be required to be sealed..., unless the cable is attached to process equipment or devices that may cause a pressure in excess of 6 inches (1493 pascals) of water to be exerted at a cable end, in which case a seal, barrier, or other means shall be provided to prevent migration of flammables into an unclassified area.

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Exception: Cables with an unbroken gas/vapor-tight continuous sheath shall be permitted to pass through a Class I, Division 2 location without seals.

(4) Cables that do not have a gas/vapor-tight continuous sheath shall be sealed at the boundary of the Division 2 and unclassified location in such a manner as to minimize the passage of gases or vapors into an unclassified location.

(FPN): The sheath mentioned in (d) and (e) above may be either metal or a nonmetallic material.

(f) Drainage.

(1) Control Equipment. Where there is a probability that liquid or other condensed vapor may be trapped within enclosures for control equipment or at any point in the raceway system, approved means shall be provided to prevent accumulation or to permit periodic draining of such liquid or condensed vapor.

(2) Motors and Generators. Where the authority having jurisdiction judges that there is a probability that liquid or condensed vapor may accumulate within motors or generators, joints and conduit systems shall be arranged to minimize entrance of liquid. If means to prevent accumulation or to permit periodic draining are judged necessary, such means shall be provided at the time of manufacture and shall be considered an integral part of the machine.

(3) Canned Pumps, Process or Service Connections, Etc. For canned pumps, process or service connections for flow, pressure, or analysis measurement, etc., that depend upon a single compression seal, diaphragm or tube to prevent flammable or combustible fluids from entering the electrical conduit system, an additional approved seal, barrier, or other means shall be provided to prevent the flammable or combustible fluid from entering the conduit system beyond the additional devices or means, if the primary seal fails.

The additional approved seal or barrier and the interconnecting enclosure shall meet the temperature and pressure conditions to which they will be subjected upon failure of the primary seal, unless other approved means are provided to accomplish the purpose above.

Drains, vents, or other devices shall be provided so that primary seal leakage will be obvious.

(FPN): See also the last paragraph of Section 500-5(b) and Fine Print Notes to Section 501-5.

502-5 Sealing, Class II, Divisions 1 and 2. Where a raceway provides communication between an enclosure that is required to be dust-ignitionproof and one that is not, suitable means shall be provided to prevent the entrance of dust into the dust-ignitionproof enclosure through the raceway. One of the following means shall be permitted: (1) a permanent and effective seal; (2) a horizontal raceway not less than 10 feet (3.05 m) long; or (3) a vertical raceway not less than 5 feet (1.52 m) long and extending downward from the dust-ignition proof enclosure.

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Where a raceway provides communication between an enclosure that is required to be dust-ignitionproof and an enclosure in an unclassified location, seals shall not be required.

Sealing fittings shall be accessible.

Exception: Seals are not required for enclosures that contain only intrinsically safe apparatus except as required by Section 501-5(f)(3).

(FPN): It is not the intent of this section to require an explosionproof seal.

See figure 4.13 for an example of sealing conduit that contains intrinsically safe circuits.

504-80 Identification. Labels required by this section shall be suitable for the environment where they are installed with consideration given to exposure to chemicals and sunlight.

(a) Terminals. Intrinsically safe circuits shall be identified at terminal and junction locations in a manner that will prevent unintentional interference with the circuits during testing and servicing.

(b) Wiring. Raceways, cable trays, and open wiring for intrinsically safe system wiring shall be identified with permanently affixed labels with the wording "Intrinsic Safety Wiring" or equivalent. The labels shall be so located as to be visible after installation and placed so that they may be readily traced through the entire length of the installation. Spacing between labels shall not be more than 25 feet (7.62 m).

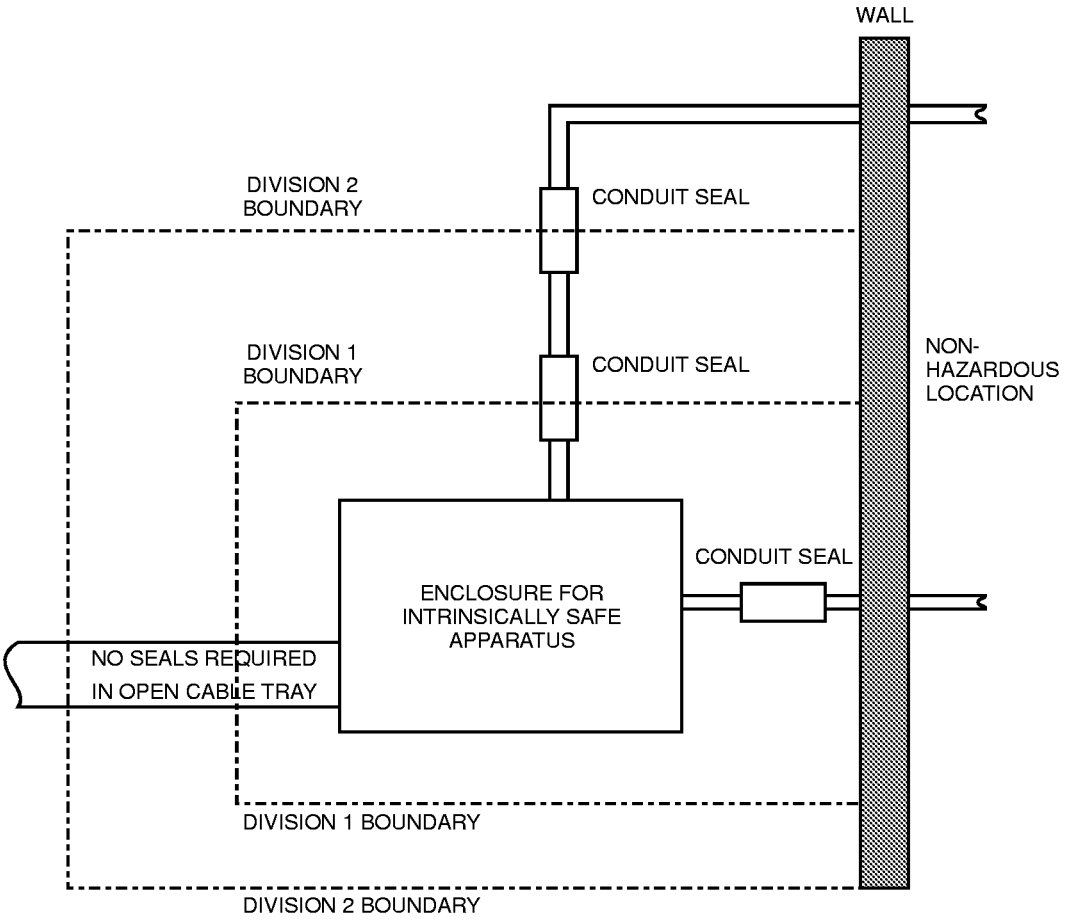
Exception: Circuits run underground shall be permitted to be identified where they become accessible after emergence from the ground.

(FPN No. 1): Wiring methods permitted in nonhazardous locations may be used for intrinsically safe systems in hazardous (classified) locations. Without labels to identify the application of the wiring, enforcement authorities cannot determine that an installation is in compliance with the Code.

(FPN No. 2): In nonhazardous locations the identification is necessary to assure that nonintrinsically safe wire will not be inadvertently added to existing raceways at a later date.

(c) Color coding. Color coding shall be permitted to identify intrinsically safe conductors where they are colored light blue and where no other conductors colored light blue are used.

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NOTES:

SEALS MUST BE LOCATED WITHIN 18 INCHES ON EITHER SIDE OF THE CLASSIFICATION CHANGE.

THERE SHALL BE NO UNION, COUPLING, BOX, OR FITTING IN THE CONDUIT BETWEEN THE CONDUIT SEAL AND THE POINT AT WHICH THE CONDUIT LEAVES THE DIVISION 1 OR DIVISION 2 LOCATION.

THE SEALS DO NOT HAVE TO BE EXPLOSIONPROOF SEALS.

Figure 4.13 — Location of conduit seals in an intrinsically safe system

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5 Guidelines for combinations of apparatus under the entity concept

5.1 General

5.1.1 The entity concept allows the user to identify acceptable combinations of intrinsically safe apparatus and associated apparatus that have not been examined as a system. Each apparatus is examined separately by a nationally recognized test laboratory (NRTL) and assigned a set of parameters called entity parameters.

5.1.2* Intrinsically safe apparatus is assigned V_{max}, I_{max}, C_i, and L_i.

5.1.3 Each channel of associated apparatus is assigned V_{oc}, I_{sc}, C_a, and L_a.

5.1.4 Combinations of channels of associated apparatus are assigned V_t, I_t, C_a, and L_a.

5.1.5 Each intrinsically safe apparatus should have a control drawing that specifies V_{max}, I_{max}, C_i, and L_i, and the terminals to which they apply. An intrinsically safe apparatus that has more than one intrinsically safe circuit may have a different set of parameters for each circuit. When this is the case, each circuit may be considered as a separate entity for connection to associated apparatus. However, the requirements of *NEC* ® Section 504-30(b) apply for separation of the circuits in the installation.

5.1.6 Each associated apparatus has a control drawing that specifies output parameters for the set of terminals to be connected to the intrinsically safe apparatus. Single-channel associated apparatus will have one set of V_{oc}, I_{sc}, C_a, and L_a parameters. Multi-channel associated apparatus will have one set of V_{oc}, I_{sc}, C_a, and L_a parameters for each channel and a separate set of V_t, I_t, C_a, and L_a parameters for combinations of channels. Systems that have more than one associated apparatus may also need a control drawing that specifies V_t, I_t, C_i, and L_i parameters of the combination of channels to be connected to the intrinsically safe apparatus. The parameters that apply to the exact interconnection must be used to assess the intrinsic safety of the system.

5.1.7 The length of cable connecting intrinsically safe equipment with associated equipment may be limited because of the energy-storing characteristics of the cable. The control drawing provides guidance in determining the maximum allowed capacitance and inductance. If the electrical parameters of the cable used are unknown, the following values may be used:

Capacitance - 60 pF/ft

Inductance - 0.20 µH/ft

5.1.8* Simple apparatus must comply with the control drawing provided with the associated apparatus.

Exception: Simple apparatus that does not interconnect intrinsically safe circuits.

5.1.9 Simple apparatus need not be listed or labeled.

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* Further information may be found in annex A.

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5.2 Assessing the intrinsic safety of combinations of intrinsically safe and associated apparatus

5.2.1 For systems that have a single-channel associated apparatus connected to only one intrinsically safe apparatus, the interconnection is intrinsically safe if:

$V_{max} \geq V_{oc}$

$I_{max} \geq I_{sc}$

$(C_i + C_{cable}) \leq C_a$

$(L_i + L_{cable}) \leq L_a$

NOTE — The capacitance and inductance of the interconnecting cable must be added to that of the intrinsically safe apparatus.

5.2.2 For systems that have more than one channel of associated apparatus connected to a single intrinsically safe apparatus, the interconnection is intrinsically safe if:

$V_{max} \geq V_t$

$I_{max} \geq I_t$

$C_a \geq (C_i + C_{cable})$

$L_a \geq (L_i + L_{cable})$

5.2.3 For systems that have a single-channel associated apparatus connected to more than one intrinsically safe apparatus, the interconnection is intrinsically safe if:

$V_{max} \geq V_{oc}$ for each intrinsically safe apparatus

$I_{max} \geq I_{sc}$ for each intrinsically safe apparatus

$C_a \geq (C_{itot} + C_{cable})$ where C_{itot} = sum of individual C_i values

$L_a \geq (L_{itot} + L_{cable})$ where L_{itot} = sum of individual L_i values

5.2.4 For systems that have more than one channel of associated apparatus connected to more than one intrinsically safe apparatus, the interconnection is intrinsically safe if:

$V_{max} \geq V_t$ for each intrinsically safe apparatus

$I_{max} \geq I_t$ for each intrinsically safe apparatus

$C_a \geq (C_{itot} + C_{cable})$ where C_{itot} = sum of individual C_i values

$L_a \geq (L_{itot} + L_{cable})$ where L_{itot} = sum of individual L_i values

5.2.5 For systems that have more than one channel of associated apparatus connected to a single intrinsically safe apparatus where separate parameters have been specified for each channel, the interconnection is intrinsically safe if, for each channel:

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V_{max} $\geq V_{oc}$

I_{max} $\geq I_{sc}$

C_a $\geq (C_i + C_{cable})$

L_a $\geq (L_i + L_{cable})$

Both associated apparatus channels must be of the same polarity.

The C_a and L_a ratings used to calculate the maximum allowed cable capacitance and inductance will be the lower value of either associated apparatus of either channel.

5.3 Intrinsically safe apparatus with more than one intrinsically safe circuit

5.3.1 Intrinsically safe apparatus with more than one intrinsically safe circuit may require special isolation between the circuits. The control drawing will specify if more than one circuit is involved and whether the circuits have to be isolated.

5.3.2 When each circuit must be isolated, the requirements of *NEC* ® clause 504-30(b) apply.

5.3.3* Maintenance should be restricted to one circuit at a time unless intrinsic safety is not impaired.

6 Maintenance and inspection

6.1 General

6.1.1 Maintenance and inspection procedures should be performed by qualified persons and should not compromise intrinsic safety.

CAUTION — ALTHOUGH INTRINSICALLY SAFE CIRCUITS ARE INHERENTLY LOW ENERGY, THEY MAY STILL PRESENT A SHOCK HAZARD BECAUSE OF THE OPERATING VOLTAGE.

6.1.2 Inspection should be performed periodically to ensure that intrinsic safety has not been compromised. Inspections should include reviewing for unauthorized modifications, corrosion, accidental damage, change of flammable materials, and the effects of aging.

6.2 Ensuring that maintenance and inspection does not compromise intrinsic safety

6.2.1 User replaceable parts of an intrinsically safe system should not be replaced with other than the manufacturer's direct equivalent.

6.2.2* Maintenance work may be performed on energized apparatus subject to the conditions detailed below:

a) Maintenance work in hazardous areas should be restricted to the following:

---

* Further information may be found in annex A.

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1) Disconnection of, and removal or replacement of, items of electrical apparatus and cabling if such action will not result in shorting of different intrinsically safe circuits.

2) Adjustment of any control that is necessary for the calibration of the electrical apparatus or system.

3) Only test instruments specified in the relevant documentation should be used.

4) Performance of other maintenance activities specifically permitted by the relevant control drawing and instruction manual.

Persons performing maintenance described above should ensure that the intrinsically safe system or self-contained intrinsically safe apparatus meets the requirements of the relevant documentation after completion of any of the work.

b) Maintenance of associated apparatus and parts of intrinsically safe circuits located in nonhazardous areas should be restricted to that described in a way such that electrical apparatus or parts of circuits remain interconnected with parts of intrinsically safe systems located in hazardous areas. Safety barrier ground connections should not be removed without first disconnecting the hazardous area circuits.

Other maintenance work on associated apparatus or parts of an intrinsically safe circuit mounted in a nonhazardous area should be performed only if the electrical apparatus or part of a circuit is disconnected from the part of the circuit located in a hazardous area.

6.2.3 The following are examples of operations not allowable without first de-energizing the intrinsically safe circuits at the associated apparatus or confirming that a flammable atmosphere is not present.

a) Disconnecting or pulling cables with multiple intrinsically safe circuits unless such action will not result in shorting of different intrinsically safe circuits — e.g., by insulating each wire termination immediately after disconnecting it from the intrinsically safe apparatus

b) Disconnecting multiple intrinsically safe circuits in the same intrinsically safe apparatus or terminal junction box unless such action will not result in shorting different intrinsically safe circuits

c) Using test equipment that is not permitted by the relevant documentation

d) Jumpering circuits or components in the intrinsically safe apparatus

6.3 Inspecting an intrinsically safe system

6.3.1 The location classification and the suitability of the intrinsically safe system for that classification should be verified. This includes verifying that the class, group, and temperature ratings of both the intrinsically safe apparatus and the associated apparatus agree with the actual classification of the location.

6.3.2 Intrinsically safe systems should be inspected to ensure that the:

a) Installation is in compliance with the documentation

b) Intrinsically safe circuits are properly separated from nonintrinsically safe circuits

c) Cable shields are grounded in accordance with the installation documentation

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d) Modifications have been authorized

e) Cables and wiring are not damaged

f) Bonding and grounding connections are tight

g) Bonding and grounding hardware is not corroded

h) Resistance of any grounding conductor, including termination resistance from shunt type associated apparatus to the grounding electrode does not exceed one ohm

i) Protection has not been defeated by bypassing

j) Printed circuit boards are clean and undamaged

6.3.3 All deficiencies should be corrected.

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Annex A — Explanatory notes

This annex is not part of ANSI/ISA-RP12.6, but is included to facilitate its use. The notes below are numbered to correspond to the related section (noted with an asterisk) in the text; therefore, the numbers do not follow a numerical sequence.

A.1.1 For formal interpretations of the requirements of *NEC* ® Article 504 and other articles of the *National Electrical Code* ®, see Article 90-5 of NFPA 70.

A.5.1.2 The values of V_{max} and I_{max} are selected by the manufacturer of the intrinsically safe apparatus to allow connection of the intrinsically safe apparatus with as wide a variety of associated apparatus as possible. V_{max} and I_{max} represent worst case associated apparatus fault conditions and do not necessarily bear any relationship to the normal operating voltage and current parameters of the intrinsically safe apparatus. V_{max} and I_{max} are limited only by the maximum voltage and current that the intrinsically safe apparatus can receive and remain intrinsically safe, based on stored energy and thermal considerations. The V_{max} and I_{max} values specified for a given intrinsically safe apparatus, taken together and compared to the ignition curves (ref. ANSI/UL 913), probably will fall in the ignition-capable area of the curve. This does not represent a problem, however, since any NRTL-approved associated apparatus must have a V_{oc} and I_{sc} combination that is not ignition-capable. For example, an intrinsically safe apparatus with low C_i and L_i values and properly rated components could realistically have a V_{max} of 45 volts and an I_{max} of 350 mA. 350 mA is well into the ignition-capable area of the ignition curve at 45 volts. However, based on the ignition curve for Groups A and B, an associated apparatus with a V_{oc} of 45 volts would have an I_{sc} of no more than 45 mA, and an associated apparatus with an I_{sc} of 350 mA would be limited to a V_{oc} of no more than 19 volts. The connection of either associated apparatus to the intrinsically safe apparatus would result in an intrinsically safe system, since in both cases, $V_{max} \geq V_{oc}$ and $I_{max} \geq I_{sc}$. Care must be taken by the user, however, to evaluate the effects of cable capacitance and inductance on the suitability of the system, and ensure that the proper operational voltage and current levels for the intrinsically safe apparatus are available from the associated apparatus selected.

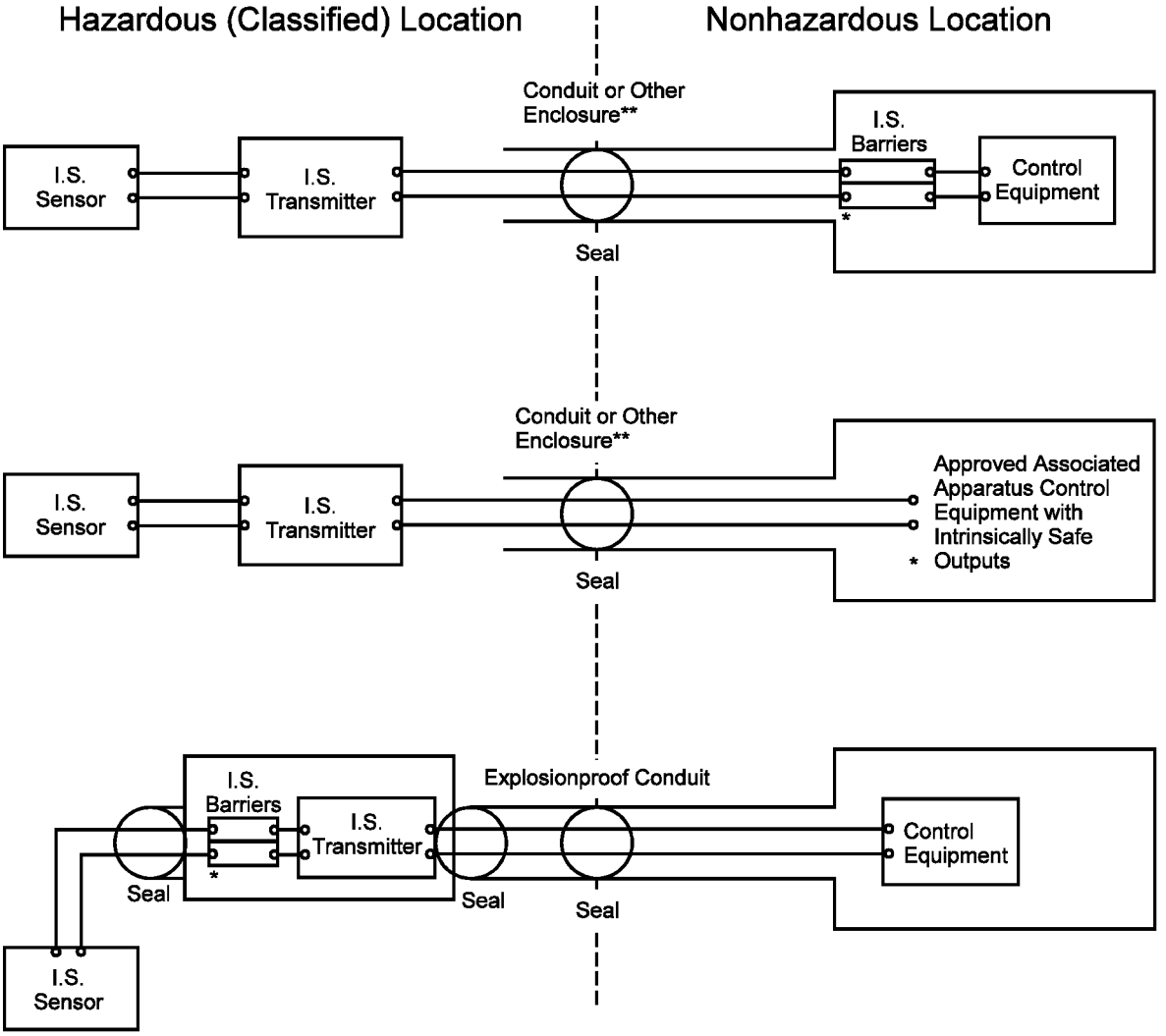
A.5.1.8 A simple apparatus may be assumed to interconnect any circuits to which it is connected. Therefore, if a simple apparatus is connected to more than one channel of associated apparatus, there must be a control drawing documenting that the combination of channels may be connected to simple apparatus.

Wiring devices such as connectors and terminal blocks may be used in intrinsically safe systems, as necessary. They are not considered as either intrinsically safe apparatus or as simple apparatus, and do not need to be shown on control drawings. The wiring devices must not compromise spacings between different intrinsically safe circuits or between intrinsically safe and non-intrinsically safe circuits.

A.5.3.3 An intrinsically safe circuit has been evaluated for the consequences of shorting, opening, or grounding the wires. However, if more than one intrinsically safe circuit is present, maintenance that intentionally or accidentally interconnects the circuits may compromise intrinsic safety.

A.6.2.2 It is preferred that maintenance and inspection be performed only when the flammable atmosphere is not present. In some cases, the flammable material may also be toxic. Ignition of the flammable atmosphere may also occur because of dropped tools, static charge, etc.

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* I.S. (Intrinsically Safe) Terminals

** Other enclosure may be shielded cable, metal-clad cable, or any mechanical or electrical protection that enforces separation of intrinsically safe circuits from circuits that are not intrinsically safe.

Figure A.1 — Various configurations of intrinsically safe systems

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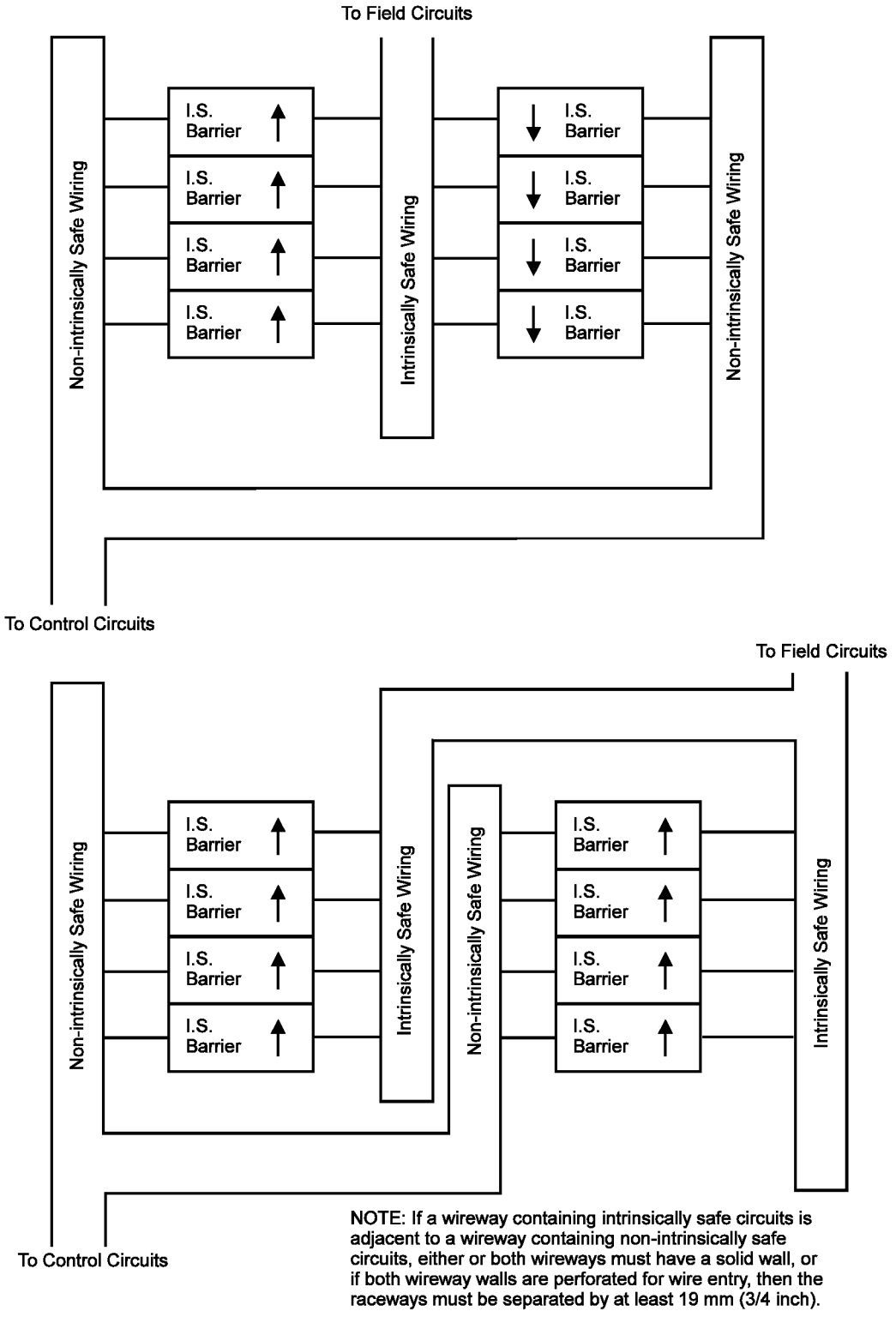


Figure A.2 — Suggested panel arrangement using separate wireways

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Annex B — Wiring in hazardous (classified) locations

This annex is not part of ISA-RP12.6, but is included to facilitate its use.

B.1 Wiring in hazardous locations

Table B.1 includes the concept of a Division 0 location, to correspond with Zone 0, as defined in the International Electrotechnical Commission Publication 79-10 (1972), "Electrical Apparatus for Explosive Gas Atmospheres, Part 10: Classification of Hazardous Areas," as an area in which an explosive atmosphere is continuously present or present for long periods. This condition is included in the definition of Division 1 in the NEC® (1993) and the CEC (1990). The concept is presented here for additional information. Table B.2 is included for the same reason.

Table B.1 — Field wiring in Class I locations [a,b]

Wiring system	Division 0		Division 1		Division 2	
	IS	NIS	IS	NIS	IS/NI	NIS
Threaded rigid metal conduit	A	Note 1 or 2	A	A	A	A
Threaded steel intermediate metal conduit	A	Note 1 or 2	A	A	A	A
Flexible metal explosionproof fitting	A	Note 1 or 2	A	A[c]	A	A
Type MI cable	A	Note 2[d]	A	A[d]	A	A
Type PLTC, MC, MV, SNM, and TC cable	A	NA	A	NA	A	A
Flexible metal conduit	A	NA	A	NA	A	A[c,e]
Liquid-tight, flexible metal conduit	A	NA	A	NA	A	A[c,e]
Electrical metallic tubing (steel)	A	NA	A	NA	A	NA
Flexible cord	A	NA	A	Note 3[f]	A	A[c,f] Notes 3,4
Any other wiring method suitable for nonhazardous locations	A	NA	A	NA	A	NA

[a] Abbreviations: IS = Intrinsically Safe; NIS = Not Intrinsically Safe; NI = Nonincendive; A = Acceptable; NA = Not Acceptable, *NEC® = National Electrical Code®* ANSI/NFPA 70-1992.

[b] See the *NEC®* for a description and use of wiring systems. Divison 0 wiring is not presently required by the *NEC®*; Divisions 1 and 2 wiring are required per the *NEC®*. Division 0 requirements are provisional recommendations only and do not represent a proposed standard.

[c] Acceptable only where flexibility is needed.

[d] Acceptable only with termination fittings approved for Class I, Division 1 locations of the proper groups.

[e] Special bonding/grounding methods for hazardous(classified) locations are required.

[f] Extra-hard-usage type with grounded conductor only acceptable.

NOTE 1 — Acceptable if entire conduit system and all enclosures are purged and pressurized using Type X purging. Acceptable if entire conduit system and all enclosures are purged and pressurized using Type Y purging, and if there are no ignition-capable parts (arcing, sparking, or high temperature) under normal operating conditions (see NFPA 496).

NOTE 2 —- Acceptable if circuit, under nomal conditions, cannot release sufficient energy to ignite hazardous atmospheric mixture when any conductor is opened, shorted to ground, or shorted to any other conductor in the same cable or raceway.

NOTE 3 — Acceptable on approved portable equipment where provisions made for cord replacement, per *NEC®* 501-11.

NOTE 4 — Acceptable on process control instruments to facilitate replacements, per *NEC®* 501-3(b) (6).

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Table B.2 — Field wiring in Class II locations [a,b]

Wiring system	Division 1		Division 2	
	IS	NIS	IS/NI	NIS
Threaded rigid metal conduit	A	A	A	A
Threaded steel intermediate metal conduit	A	A	A	A
Flexible metal explosionproof fitting	A	A[c]	A	A[c]
Type MI cable	A[d]	A[e]	A	A
Type MC and SNM cable	A[d]	NA	A	A
Type PLTC and TC cable	A[d]	NA	A	A[f]
Flexible metal conduit	A[d]	NA	A	NA
Liquid-tight, flexible metal conduit	A[d]	A[c,d,g]	A	A[c,d,g]
Flexible cord	A[d]	A[c,d,h]	A	A[c,d,h]
Dust-tight wireways and raceways	A	NA	A	A
Any other wiring method suitable for nonhazardous locations	A[d]	NA	A	NA
Electrical metallic tubing	A[d]	NA	A	A

[a] Abbreviations: IS = Intrinsically Safe; NIS = Not Intrinsically Safe nor nonincendive; A = Acceptable; NA = Not Acceptable; NI = Nonincendive, *NEC®* = *National Electrical Code®* ANSI/NFPA 70-1992.

[b] See the *NEC®* for description and use of wiring systems.

[c] Acceptable only where flexibility is needed.

[d] Acceptable only with dust-tight seals at both ends when electrically conductive dusts will be present.

[e] Acceptable only with termination fittings approved for Class II, Division 1 locations of the proper groups.

[f] Acceptable in ventilated channel-type cable trays in a single layer for a space not less than the larger cable diameter between adjacent cables.

[g] Special bonding/grounding methods for hazardous (classified) locations are required.

[h] Extra-hard-usage type with grounded conductor only acceptable.

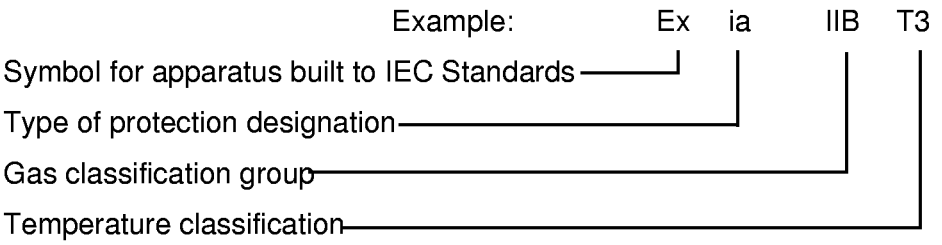
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Annex C — Contents of foreign marking labels for apparatus for use in hazardous (classified) locations

This annex is not part of ANSI/ISA-RP12.6, but is included to facilitate its use.

C.1 Explanation of label

IEC (International Electrotechnical Commission) standards require the following marking, which is similar to the CENELEC (European) recommendations, except that the CENELEC "EEx" symbol replaces the IEC symbol "Ex" symbol.



In addition, the testing station and the number of the test certificate or certificate of conformity are indicated.

C.1.1 Type of protection designation

d - Flameproof enclosure

e - Increased safety

ia - Intrinsic safety (Zone 0)

ib - Intrinsic safety (Zone 1)

h - Hermetically sealed

m - Encapsulation

n - Nonsparking

o - Oil immersion

p - Pressurized enclosure

q - Powder-filled

s - Special protection

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C.1.2 Gas classification group

IIA similar to *NEC*® Group D

IIB similar to *NEC*® Group C

IIB + H_2 similar to *NEC*® Group B

IIC similar to the combined *NEC*® Groups A & B + carbon disulfide

C.1.3 Temperature classification

The number values correspond to the Identification Numbers in Table 500-3(b) of the *NEC*® (1993) and to the Temperature codes in Rule 18-052(2) of the CEC (1990).

T1 = 450°C (842°F)

T2 = 300°C (572°F)

T3 = 200°C (392°F)

T4 = 135°C (275°F)

T5 = 100°C (212°F)

T6 = 85°C (185°F)

C.2 Comparison of IEC "Zones" to North American "Divisions" and the types of protection accepted

IEC		North America	
Zone 0	Intrinsically safe apparatus of category ia or other apparatus specifically approved for Zone 0	Class I, Division 1	Some users recognize the Zone 0 principle without using the name and would only install apparatus suitable for Zone 0 operation in such areas.
Zone 1	Apparatus with type(s) protection: - 'd' flameproof enclosure - 'e' increased safety - 'i' intrinsic safety (ia and ib) - 'm' encapsulation - 'o' oil immersion - 'p' pressurized apparatus - 'q' powder filling - 's' special protection		Apparatus with type(s) of protection: - explosionproof enclosures - pressurization - intrinsic safety - oil immersion
Zone 2	All equipment certified for Zone 0 or 1 Apparatus with type of protection: - 'n' nonsparking/nonincendive	Class I, Division 2	All equipment certified for Division 1 or 2 Apparatus with type of protection: - nonincendive (ANSI/ISA-S12.12) Apparatus without make-and-break or sliding contacts in "general purpose" enclosures, ANSI/NFPA 70 Section 501- 3(b)(2) Exception.*

* See ISA-S12.1 for a list of standards related to each type of protection.

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Annex D — References

CANADIAN STANDARDS ASSOCIATION (CSA)

C22.1, Part 1 Canadian Electrical Code (CEC), 1990: Safety Standards for Electrical Installations

Available from: **CSA**
178 Rexdale Blvd.,
Etobicoke, Ontario M9W 1R3
Canada
Tel: (416) 747-4044
Telex: 06 989344

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 79-10 Electrical Apparatus for Explosive Gas Atmospheres, Part 10: Classification of Hazardous Areas, 1972

Available from: **American National Standards Institute (ANSI)**
11 West 42nd Street
New York, NY 10036
Tel: (212) 642-4900

ISA

ISA-S12.1-1991 Definitions and Information Pertaining to Electrical Instruments in Hazardous (Classified) Locations

ANSI/ISA-S12.12-1994 Nonincendive Electrical Equipment for Use in Class I and II, Division 2 and Class III, Divisions 1 and 2 Hazardous (Classified) Locations

Available from: **ISA**
67 Alexander Drive
P.O. Box 12277
Research Triangle Park, NC 27709
Tel: (919) 549-8411

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

ANSI/NFPA 70 *National Electrical Code®*, 1993

Available from: **NFPA**
Batterymarch Park
Quincy, MA 02269
Tel: (617) 770-3000

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UNDERWRITERS LABORATORIES, INC. (UL)

ANSI/UL 913 Intrinsically Safe Apparatus and Associated Apparatus for Use in Class I, II, and III, Division I, Hazardous (Classified) Locations

Available from: **UL**
333 Pfingsten Road
Northbrook, IL 60062
Tel: (708) 272-8800
Fax: (708) 272-8129

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